

10 — Upgrade & Rollback

Audience: whoever runs a production deploy. **Companion docs:** [09 Backup & restore](#) · [02 Deployment](#) · [13 Operations](#) **The tool:** `deploy/prism-deploy.sh`

1. The five rules the script exists to enforce

Read these before the commands. Every design decision below follows from one of them.

- 1. Nothing is touched until a backup exists.** The database dump and the three secret locations are captured, verified non-empty, and written **outside the release tree** before a single container is stopped.
- 2. The new release is built before the old one is disturbed.** A build failure — a bad Dockerfile, a full disk, a network blip pulling a base image — leaves the running stack untouched and exits non-zero. The swap happens only once every image exists.
- 3. Failure rolls itself back.** If the stack does not come back healthy inside `HEALTH_TIMEOUT`, the script restores the previous tree *and images* itself rather than leaving a half-deployed platform for someone to find.
- 4. The project name is pinned.** Compose derives it from the compose file's directory, so it is stable across tree swaps — but it is passed explicitly anyway, because a drifted project name would mean new containers pointed at new, *empty* volumes, and the database would look wiped when it is merely orphaned.
- 5. Volumes are never removed.** `down -v` appears nowhere in the file, and `down` itself is not used — services are recreated in place.

And one deliberate non-goal:

It never restores the database automatically. An upgrade that fails is a code problem; the data is fine, and silently rewinding it would destroy work the desk did between the backup and the failure. `rollback --with-db` exists for the rare case where you decide otherwise, and it dumps the current state first.

2. Commands

```
./prism-deploy.sh upgrade ~/prism-b423fa2.zip # backup → build → swap → verify
./prism-deploy.sh upgrade ~/prism.tar.gz      # .zip .tar .tar.gz .tgz all accepted
./prism-deploy.sh rollback                    # back to the previous release
./prism-deploy.sh rollback --with-db          # ...and the database with it
./prism-deploy.sh status                      # what is live, what can be rolled back to
./prism-deploy.sh backup                      # a dump + secrets snapshot, nothing else
./prism-deploy.sh verify                      # health-check the running stack
./prism-deploy.sh restore-db <file.sql.gz>    # a specific dump, on purpose
```

Where to run it from

The script must live **outside** the tree it swaps — a copy inside the release is replaced mid-run. Both placements are resolved automatically:

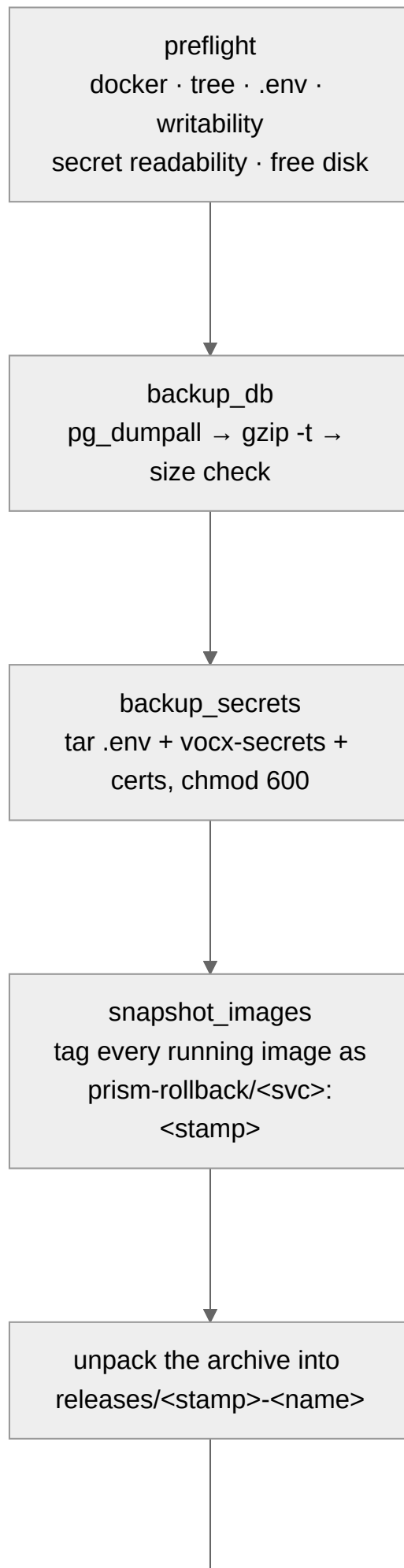
```
$PRISM_ROOT/prism-deploy.sh      ← the recommended home, and where `upgrade` installs it
$CHECKOUT/depoy/prism-deploy.sh ← works too; ROOT resolves two levels up
```

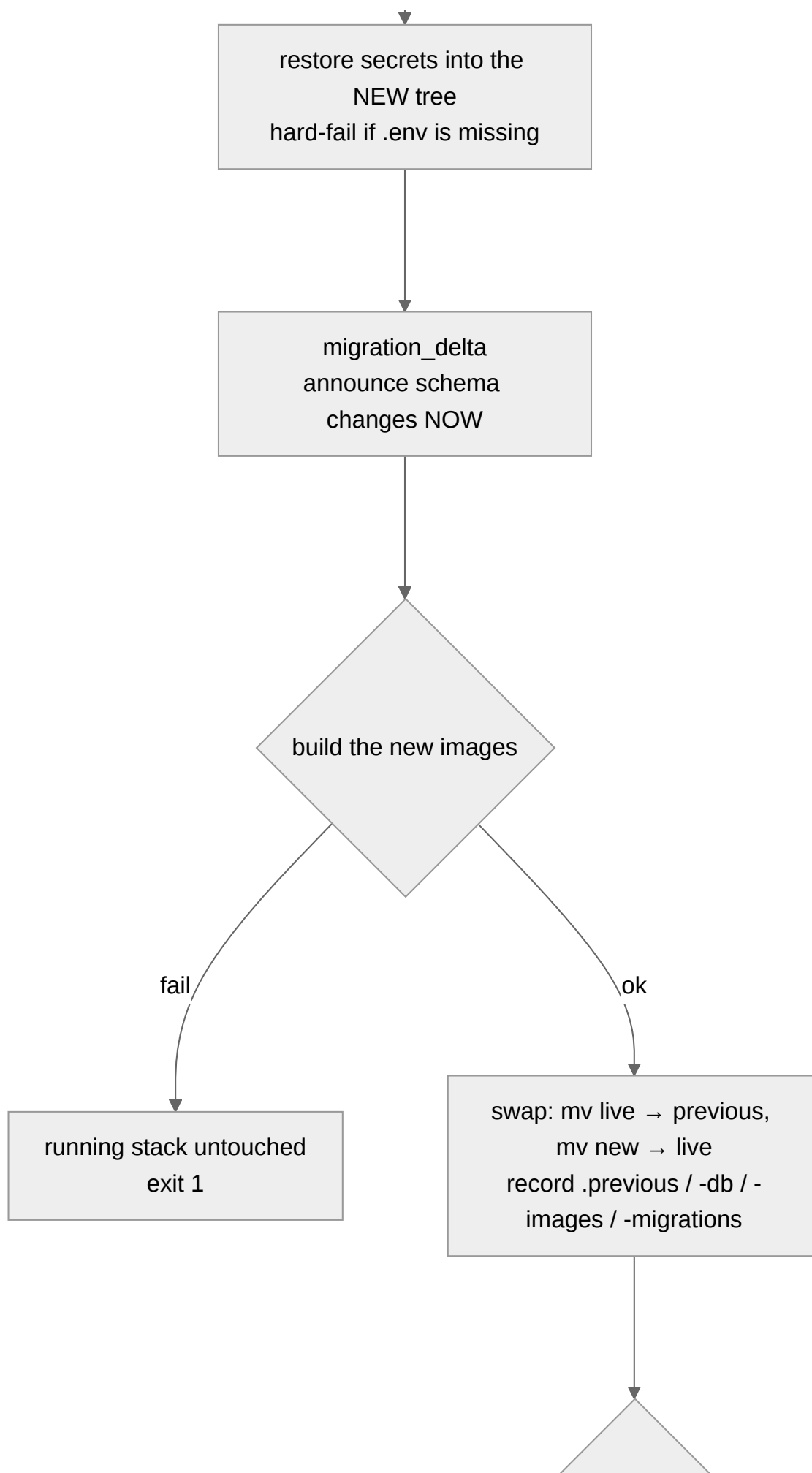
`PRISM_ROOT` overrides the detection. On Evam production, `$PRISM_ROOT = /home/ubuntu/aug_11`.

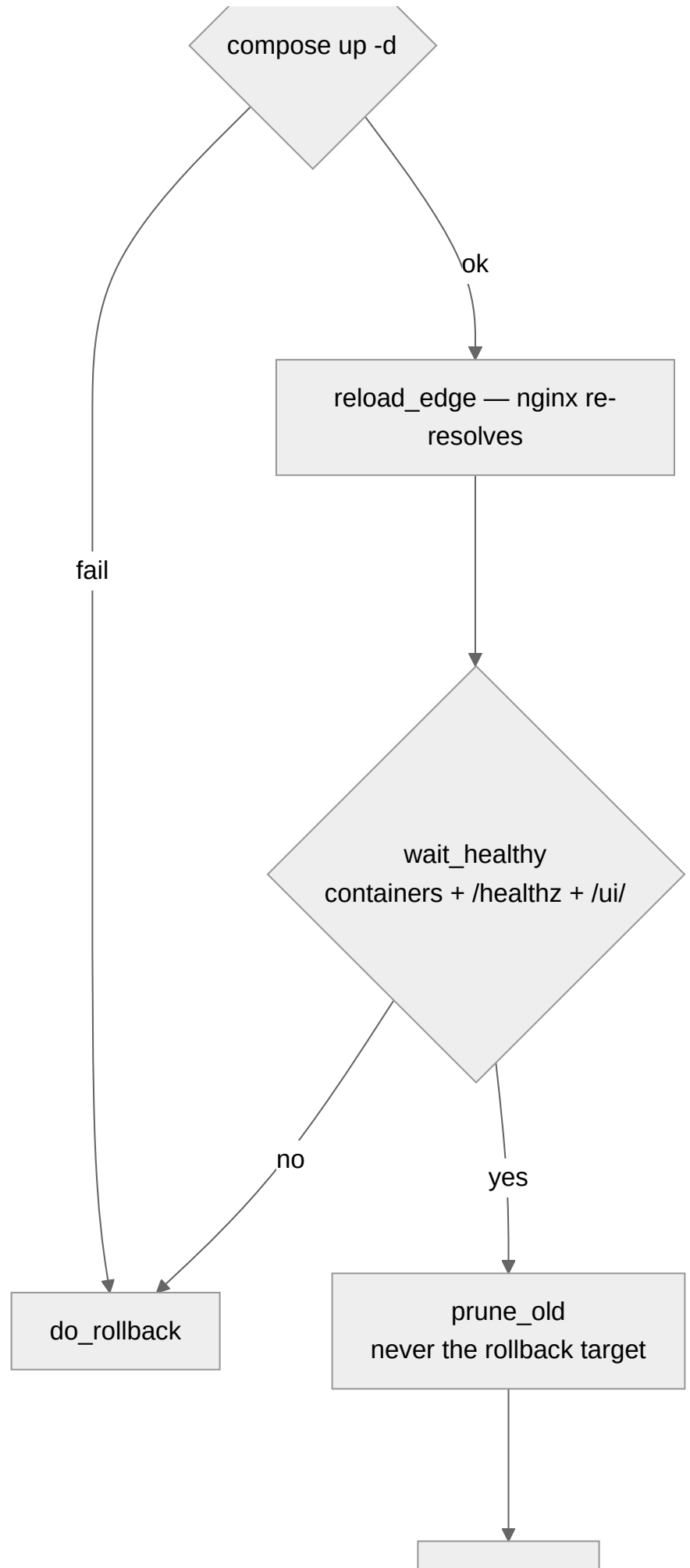
Configuration

Variable	Default	Meaning
<code>PRISM_ROOT</code>	auto-detected	The deployment root
<code>PRISM_PROFILES</code>	<code>sso backup</code>	Compose profiles this deployment runs with
<code>PRISM_KEEP_RELEASES</code>	3	Old trees retained (the rollback target is <i>never</i> counted)
<code>PRISM_KEEP_BACKUPS</code>	10	Dumps retained
<code>PRISM_HEALTH_TIMEOUT</code>	180	Seconds to become healthy after a swap
<code>PRISM_MIN_FREE_GB</code>	8	Refuse to start without this much free disk

3. What `upgrade` does, step by step







complete

Preflight — the checks, and why each exists

Check	The failure it prevents
docker present, daemon reachable	An obvious failure discovered halfway through
<code>\$LIVE</code> is a PRISM tree with a <code>.env</code>	Upgrading the wrong directory
<code>\$ROOT</code> , <code>backups/</code> , <code>releases/</code> are writable	"Swapping releases is two <code>mv</code> s inside <code>\$ROOT</code> , so write permission there is as load-bearing as read permission on the secrets — and finding out at the swap means a full build was spent first."
Every secret file is readable by the current user	A root-owned <code>tls.key</code> under an <code>ubuntu</code> run. Checked first , named precisely, with both remedies printed — because "a snapshot that silently skipped a key would restore a tree that cannot serve TLS."
<code>≥ MIN_FREE_GB</code> free	A build that fills the disk mid-way

If the secrets are unreadable you get the file names and two exact commands:

```
sudo ./prism-deploy.sh upgrade ~/prism-b423fa2.zip
# ...or hand ownership over once:
sudo chown -R ubuntu:ubuntu ~/aug_11/prism/deploy/nginx/certs ~/aug_11/prism/deploy/vocx-secrets
```

The image snapshot — why rollback is fast

```
docker tag "$img" "prism-rollback/$svc:$STAMP"
```

"A tagged image is not dangling, so `docker image prune` leaves it alone — which is the whole point: the rollback path must not depend on an untagged layer nobody promised to keep."

Rollback re-points the compose image names at these tags and runs `up -d --no-build`. If a snapshot image has gone missing, it falls back to rebuilding — slower, same result.

4. Schema changes

`migration_delta()` diffs the Alembic `versions/` directories of the live and new trees, for **both** the register and access services, and prints the result **before anything moves**:

```
► This release carries SCHEMA CHANGES
  services/register: 20260812_add_reconciliation_items.py
```

They apply automatically when the services start (alembic upgrade head).
A plain rollback does NOT undo them — it puts the old code on the new schema. That is safe for an added column and unsafe for a dropped one.
If you need the old schema back too: `./prism-deploy.sh rollback --with-db`

Or, when there are none:

No schema changes in this release – rollback is fully symmetric.

This is the single most important line of output in an upgrade. It tells you, before the build, whether your rollback is symmetric.

Change type	Plain <code>rollback</code>	Verdict
Added table / added nullable column	Old code ignores it	Safe
Added NOT NULL column with a default	Old code ignores it	Safe
Renamed column	Old code queries a name that is gone	Needs <code>--with-db</code>
Dropped column / table	Old code queries something gone	Needs <code>--with-db</code>
Changed a type or constraint	Depends	Read the migration

The delta is recorded in `releases/.previous-migrations`, and `do_rollback` re-prints the warning at rollback time if you did not pass `--with-db`.

5. The health gate — both doors

```
health_once() {
  # every container that declares a healthcheck must not be unhealthy/starting
  (( unhealthy == 0 )) || return 1
  curl -sf -m 10 "http://127.0.0.1:${port}/healthz" >/dev/null || return 1
  curl -sf -m 10 -o /dev/null "http://127.0.0.1:${port}/ui/" || return 1
}
```

The second curl was added after a real production incident:

"`/healthz` proves the GATEWAY lane. It says nothing about the UI, which nginx reaches through a different upstream — and a stranded `ui` upstream passed this gate happily while every page in the browser answered 502."

Container state alone would call a stack healthy that answers 502 at the door. Both lanes must answer.

6. The edge reload — the 502 that looks like a healthy stack

```
reload_edge() {
  dc "$LIVE" ps --status running --services | grep -qx nginx || return 0
  dc "$LIVE" exec -T nginx nginx -s reload || dc "$LIVE" restart nginx
}
```

The root cause, in the script's own words:

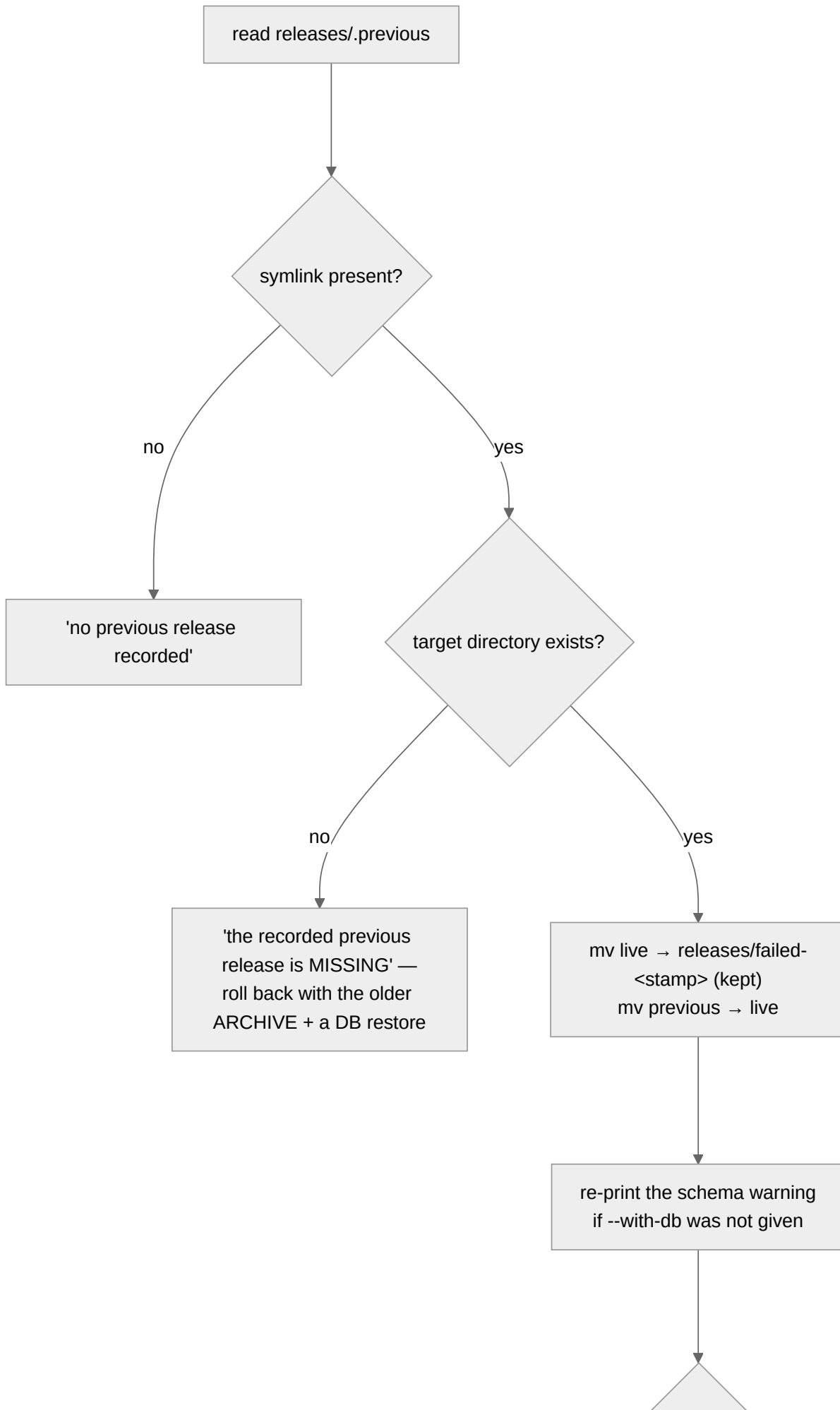
"nginx.conf declares static upstreams (`upstream ui { server ui:80; }`), and nginx resolves those names ONCE at startup, for the life of the worker. An upgrade recreates the ui and gateway containers, Docker hands them new addresses, and nginx — whose own image did not change, so it was never recreated — keeps dialling the old ones. Every page then 502s with 'connect() failed (111: Connection refused)' against a container that no longer exists, while every container reports perfectly healthy."

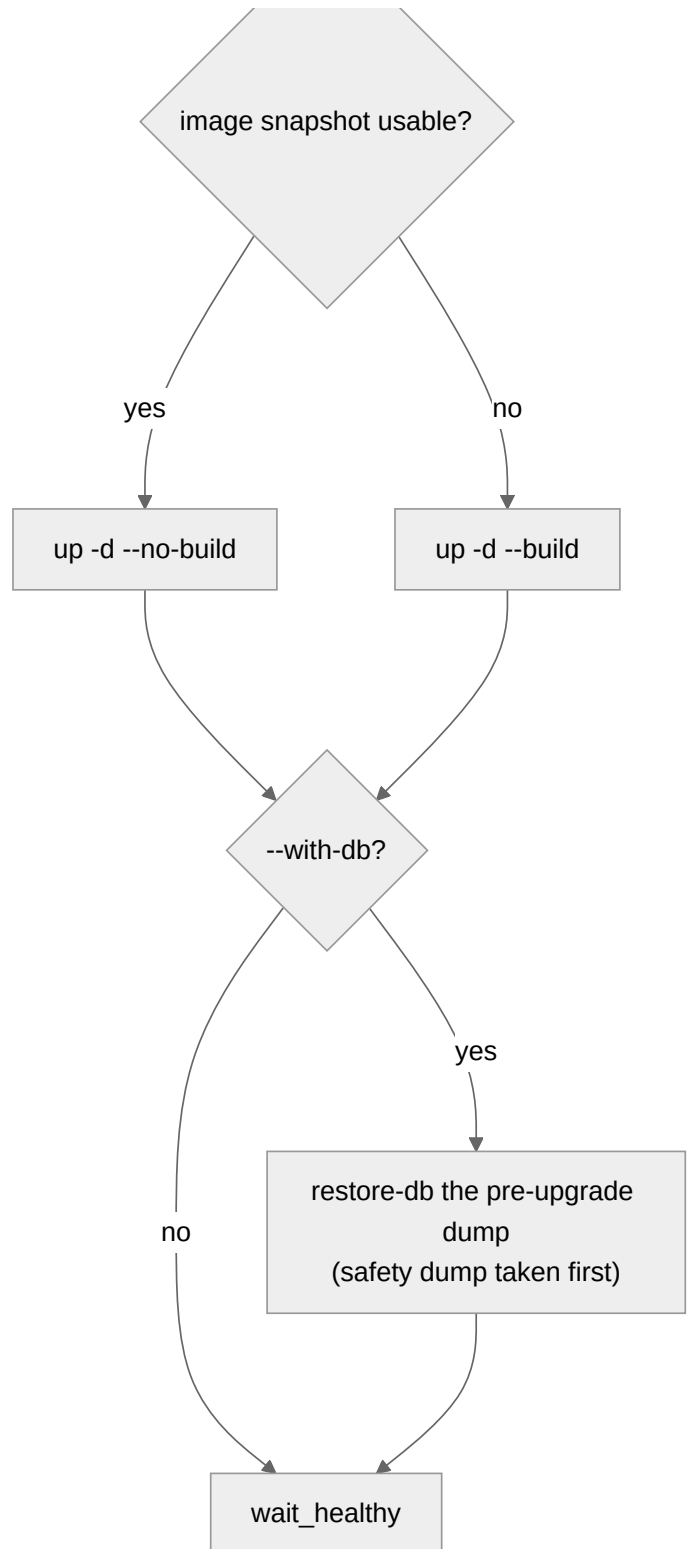
A reload re-reads the config and re-resolves without dropping a connection; a restart is the fallback if the reload is refused.

If you ever recreate `ui` or `gateway` by hand, reload nginx yourself.

7. Rollback

```
./prism-deploy.sh rollback          # code only – the recommended default
./prism-deploy.sh rollback --with-db # code AND data
```





Two details that matter:

- **The failed release is kept**, at `releases/failed-<stamp>`, for inspection. It is not deleted.
- **A recorded-but-missing target is a distinct message.** *"The tree is gone, so the way back is the release archive plus a database restore, not this command."* This exists because `prune_old` used to delete the very tree `.previous` pointed at — found by running five upgrade/rollback cycles at `KEEP_RELEASES=1`, and fixed:

```
prune_old() {  
  # THE ROLLBACK TARGET IS NEVER PRUNED. ... the failure is silent until the day someone  
  # needs it, when rollback reports "nothing to roll back to" while the symlink sits  
  # there pointing at a directory that no longer exists.  
  local keep; keep="$(readlink -f "$RELEASES/.previous" 2>/dev/null || true)"  
  ...  
}
```

8. A production upgrade, start to finish

```
# 0. On your machine: build the release archive and PROVE it carries no secrets
zip -r ~/prism-b423fa2.zip prism/ -x '*/node_modules/*' '*/.git/*'
unzip -l ~/prism-b423fa2.zip | \
    grep -Ei "client_secret|token\.json|vocx-secrets/\.json|vocx_tokens|compose/\.env"
# exit 1 (no output) = clean. ANY output = do not upload that file.

# 1. Copy it up
scp ~/prism-b423fa2.zip ubuntu@prod:~/

# 2. On the box – see where you stand first
cd ~/aug_11
./prism-deploy.sh status

# 3. Upgrade
./prism-deploy.sh upgrade ~/prism-b423fa2.zip

# 4. Read the SCHEMA CHANGES section of the output. Note the db backup path.

# 5. Verify beyond the script's own gate
./prism-deploy.sh verify
# then in a browser: sign in, open Deals, record a short VocX capture

# 6. Hard-refresh the browser (Ctrl+Shift+R) – the UI is a cached bundle
```

If step 5 looks wrong:

```
./prism-deploy.sh rollback          # code only – start here
./prism-deploy.sh rollback --with-db # only if the release dropped or renamed schema
```

9. Helm upgrades

```
helm upgrade --install prism deploy/helm/prism -n prism -f prod-values.yaml
helm history prism -n prism
helm rollback prism <revision> -n prism
```

Differences from the compose path that will surprise you:

- **Helm does not back up your database.** Take a dump yourself first. There is no rule 1 here — you are rule 1.
- **Migrations still run on service start.** The same asymmetry applies: `helm rollback` restores manifests, not schema.
- **stt has limits.cpu: 2**, so `STT_CPU_THREADS=2` is correct there. On compose there is **no** CPU limit and 2 may be a throttle — derive it from `nproc`.
- **Two Ingresses.** The slow-path Ingress carries the long timeouts; set `gateway.ingress.slowPaths` from a **values file**, not `--set` (`--set` mis-parses the list).

10. Failure playbook

Symptom	Cause	Do this
"cannot write to ... as ubuntu"	Ownership of <code>\$ROOT</code>	<code>sudo ./prism-deploy.sh ...</code> , or <code>chown</code> the root
"these secret files cannot be read"	root-owned <code>tls.key</code>	Use <code>sudo</code> , or <code>chown</code> the two directories it names
"only 3G free"	Disk	Clear old releases/images; <code>docker image prune</code> (tagged rollback images are safe)
Build fails	Code or network	Nothing was touched. Fix and re-run
"did not become healthy" then auto-rollback	New release is broken	Read <code>backups/deploy-<stamp>.log</code> ; the failed tree is at <code>releases/failed-<stamp></code>
Everything healthy but the browser 502s	Stale nginx upstream	<code>docker compose exec nginx nginx -s reload</code>
UI shows the old version	Cached bundle	Ctrl+Shift+R
Rollback says "nothing to roll back to"	Never upgraded with this script	Deploy the older archive with <code>upgrade</code>
Rollback says the previous release is "MISSING"	The tree was removed	Re-run <code>upgrade</code> with the older archive, then <code>restore-db</code> if needed
Database looks empty after an upgrade	Compose project name drifted → new empty volumes	Check <code>docker volume ls</code> ; the data is in <code>compose_pgdata</code> . Rule 4 exists for this

11. Hard-won lessons encoded in this script

Each of these was a real failure, and each is now a line of code. They are listed so nobody re-learns them.

Lesson	Where it lives
A health check on one lane passes while the other is broken	the <code>/ui/</code> probe in <code>health_once</code>
nginx caches upstream IPs across a container swap	<code>reload_edge</code>
Pruning by age deletes your own rollback target	the <code>keep</code> exclusion in <code>prune_old</code>
A failed <code>exec 9></code> redirection kills a non-interactive shell silently	the writability test runs <i>before</i> <code>exec 9>"\$LOCK"</code>
<code>say()</code> pipes through <code>tee</code> , so an unwritable log kills the script on line one	<code>LOG</code> falls back to <code>/dev/null</code>
<code>local tree="\$1" out=(-f "\$tree/...")</code> evaluates the array before <code>tree</code> is assigned	<code>compose_files()</code> uses two separate <code>local</code> statements
Discovering unreadable secrets <i>after</i> the database dump wastes the operator's nerve	the readability check is in preflight, first
A dump that exists is not a dump that restores	<code>gzip -t</code> + a 100 KB floor